

Readiness & Body Literacy

for Adolescent Female Athletes (11–18)

Master approach & tracking model — v1.0 (pre-storyboard reference)

An individualised, interoceptive model that helps girls learn what their body's signals mean, so they know when to push, rest or recover — and can tell an adult.

Purpose of this document. To verbalise the full approach — the vision, the science, the design principles, exactly what we track and how, the daily rhythm, and the guardrails — in one place, so the team, clinical advisors and developers share one reference before we storyboard the screens.

What this document covers

- 1. Vision & the problem
- 2. Scientific foundations
- 3. Guiding principles (the design DNA)
- 4. What we track — signal domains, indicators, body map, sensation lexicon, cycle
- 5. The daily rhythm — pre/post check-ins
- 6. Wearable tiers
- 7. Measuring body literacy — the metric
- 8. Defining 'normal' & the individualised model
- 9. Longitudinal data → patterns, insights & real-time research
- 10. Guardrails & safeguarding
- 11. Outputs & communicating with adults
- 12. Deliberate exclusions
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1 · Vision & the problem

The vision. Most sports technology measures the athlete *from the outside* and tells her what to do. We do the opposite: we teach an adolescent girl to read her own body — to notice its signals, learn what is normal *for her*, and use that to decide when to push, when to rest and when to recover, and to communicate it clearly to the adults around her.

The problem. Adolescent female athletes are under-served and under-researched. They carry high overuse-injury and burnout risk; menstrual dysfunction and low energy availability (RED-S) raise injury and long-term bone-health risk; and monitoring tools are mostly built on adult men. They are also at an age where body awareness (interoception) is still developing and body-image sensitivity is high. The result is girls who often can't yet tell 'good' training discomfort from a warning sign, and feel awkward putting either into words to a coach.

The opportunity. Adolescence is a sensitive developmental window for building interoception. A simple, supportive tool used consistently in these years can teach a durable life skill — body literacy — while generating the longitudinal data that makes each girl's 'normal' visible. And because that data is consented and de-identified, it also builds, in aggregate, the female-athlete evidence base the field currently lacks (see §9).

2 · Scientific foundations

The model rests on three evidence sources, each contributing a pillar:

Source	What it contributes
Welsh et al. (2026) — Interoceptive Intelligence in Sports	The core mechanism: interoception is trainable by shrinking the gap between predicted and actual bodily states (prediction error). Wearables must be calibration tools, not verdicts, or they cause dependence and nocebo effects. Adolescence is a plasticity window.
Temm, Standing & Best (2022) — Monitoring in Female Youth Athletes	The feasibility rules: simple, low-cost, individualised, sustainable, or adherence fails. Green-standard tools (sRPE, short wellness items, cycle counting-back). RED-S and overuse are the dominant risks. Youth under-report and mis-rate.
Beato et al. (2024) — Readiness in Female Football	Readiness = objective + subjective, assessed against the athlete's own baseline. HRV is highly individual. Menstrual dysfunction warrants referral. Many single-item tools aren't yet validated — check reliability before trusting them.

3 · Guiding principles (the design DNA)

- **Self-report is the backbone; the wearable is a bonus.** The app is fully useful with no device; a simple band only enhances it.
- **Interoceptive calibration, not monitoring.** She predicts, then checks; the shrinking prediction error is the point.
- **Simplicity buys adherence.** Tiny daily burden, few validated items, or the data becomes invalid.
- **Individualised 'normal'.** Everything is judged within-person over time, never against a population norm.
- **Body literacy, not surveillance.** Curious, supportive framing that builds a skill she owns and an internal locus of control.
- **Food-safe.** We track energy and thirst, never hunger, fullness or food, given age-related food sensitivity.
- **Privacy, agency & safeguarding first.** Her data is hers; sharing is opt-in; intimate and menstrual data are specially protected.
- **Evidence humility.** No false-precision thresholds; within-athlete first; clinician oversight for anything health-related.
- **Built to be spoken.** Every output should help her communicate her state to an adult.

4 • What we track

4.1 The six signal domains and how each is captured

A body map is spatial; most bodily states are not. Each domain is captured by the method that fits it:

Signal domain	Capture method	Notes
Legs & body (musculoskeletal)	Body map — primary	Region + sensation + intensity + behaviour tag.
Breath & heart (effort/arousal)	Scalar; map optional	Effort + 'nerves' 1–5; chest/throat taps allowed, not forced.
Energy & thirst	Scalar 1–5	Not localisable. No food/hunger.
Sleep & recovery	Scalar 1–5 + rough hours	Not localisable; recalled at pre-check.
Focus & mood	Emoji / 1–5	Not localisable.
Cycle (female layer)	Calendar + discreet map	Opt-in, privacy-guarded; context + RED-S safety-net.

4.2 The core indicators (grounded in Temm & Beato)

A small, validated, low-cost set — essentially the Hooper index plus load and mood — captured on a 1–5 scale inside the same check-ins. Kept few on purpose; more items erode youth adherence.

Indicator	Type	Why (evidence)
Sleep (quality + hours)	Physio	Core recovery marker; Hooper (Temm, Beato)
Fatigue / freshness	Physio	Central, multifactorial readiness signal (Temm)
Muscle soreness	Physio	DOMS/soreness + injury awareness (Temm, Beato)
Session-RPE (load)	Physio	Validated youth 'green-standard' load measure (Temm)
Stress (incl. school/exams)	Psych	Youth stress is cyclical; raises illness/injury risk (Temm)
Mood / motivation	Psych	Subjective readiness & wellbeing marker (Beato)
Resting morning HR	Physio (optional)	Autonomic readiness — individual baseline only (Beato)

4.3 The sensation lexicon (clinically mapped, teen-safe)

On the body map she taps **sensations, not just pain** — neutral and positive options included, to build literacy and avoid hypervigilance. Each maps behind the scenes to a clinical lean; she describes, the app never diagnoses. Every tap also gets an intensity (1–5) and a behaviour tag (*eases with warm-up / worse with load / there at rest*).

She taps...	Clinical category	Lean / action
Strong / warm / light / fine	Normal / positive	Neutral — reinforces good signals
Sore / achy	DOMS-type soreness	Usually adaptive; watch timing & bilaterality
Tight / stiff	Muscle/fascia tension	Monitor; mobility, not alarm
Sharp / stabbing	Acute nociceptive	Caution — especially if new/focal
Burning / tingling / numb	Possible neural	Caution — flag if it persists
Unstable / gives way	Joint instability	Red flag (knee/ankle common in girls)
Swollen / hot	Inflammation	Red flag → assess

4.4 The cycle & RED-S layer

Tracked by simple bleeding counting-back (the youth green standard), with a contraception question up front (naturally cycling / on hormonal contraception / dysfunction-suspected), because ~half of athletes use contraception that changes the signal. Cycle is used as gentle *context* — not a precision normaliser, since cycles are often irregular in the first years after menarche. Absent or irregular periods, or low energy, route quietly to a RED-S / low-energy-availability check and a prompt to see a clinician — never a food conversation.

5 · The daily rhythm — two check-ins

Two check-ins on a training day, hard-capped. Predict & sense before; look back, sense, learn & recover after. Each ≤ 60–90 seconds.

Pre-training (predict → then sense)

1. **Predict the session.** How she expects it to go, how she'll approach it (push / steady / cautious), and what she might tell her coach. This forecast is compared after training.
2. **Then sense & locate.** Body map + quick scalars, framed to invite surprise: 'did anything surprise you vs what you expected?' The surprise is logged as data.
3. **Auto: overnight recovery.** Sleep recall + resting HR (if she has a band).

Why predict first: committing a prediction before observing is what trains interoception (prediction-error learning).

The safeguard: score accuracy against the post-training outcome, not the primed pre-scan, and keep prediction a low-stakes game to avoid negative self-fulfilling forecasts.

Post-training (look back → sense → learn → recover)

4. **Look back.** Was the morning read right? This pre-vs-actual gap is the headline session prediction error.
5. **Sense & locate now.** Captures acute/effort signals — not DOMS yet (that appears next morning).
6. **Learn + load.** Note the insight; one session-RPE tap.
7. **Recover + (optional) communicate.** Food-neutral recovery nudge; optional prompt to tell an adult.

Special days & safety

- **PE + training same day:** still two checks — pre before the first activity, post after the last; tag the bouts. Never scale checks with bouts.
- **PE-only day:** a lighter pre/post pair (PE counts as load).
- **Rest day:** one optional reflection (recovery, lingering signals, cycle/energy).
- **Anytime:** acute red-flag capture (sharp pain, giving-way) is always one tap away.

You don't lose delayed soreness: tomorrow's pre-check is today's ~24 h DOMS readout. Two checks yield three learning signals — micro (predict→sense surprise), macro (pre→post session error), and cross-day (soreness evolution).

6 · Wearable tiers

Tier	Inputs	What it adds
Tier 0 — Everyone (free)	Check-ins + body map + cycle log	Full body-literacy loop; awareness + insight
Tier 1 — Simple wearable	+ resting HR & sleep	A coarse recovered / run-down context flag vs her own baseline
Tier 2 — HRV-capable	+ morning HRV (RMSSD)	Optional autonomic layer — individual-level only, never the decider

7 · Measuring body literacy — the metric

Three facets (Welsh, via Garfinkel), mapped to whether she has a device:

- **Awareness — did she notice?** No device needed.
- **Insight — how sure was she, and how right?** No device needed; core free-tier signal.
- **Accuracy — felt vs measured.** Wearable only; shown as 'how close', never a good/bad score.

Headline metric: prediction error, trending down. A shrinking gap between predicted and actual (felt, or measured) is the operational definition of growing readiness and body literacy — and it works with or without a device.

8 · Defining 'normal' & the individualised model

'Normal' is her own recurring pattern, built over time and phase-aware. From her accumulating predict/sense history the app assembles a personal *fingerprint*: her typical signals per site and per domain, how they shift across her cycle, and how accurately she reads each. The model flags **deviations from her fingerprint** — a new focal site, a signal that doesn't track load, one that persists or escalates — not absolute values.

Modelling notes. Expect sparse, high-dimensional data (regions × sensations × intensity, mostly empty per girl). Use hierarchical / partial-pooling methods that borrow strength across regions and athletes to handle cold-start, rather than a separate model per body part. Couple every signal with load. Start descriptive and within-athlete; validate before any threshold drives a decision.

9 · Longitudinal data → patterns, insights & real-time research

Within-athlete (she and her support team)

- Prediction-error trend per domain — is she getting better at reading her body?
- Her fingerprint and how it moves across the menstrual cycle (symptom map).
- Readiness patterns and load–recovery coupling (RPE vs felt/actual recovery).
- Quiet RED-S / low-energy safety-net from the energy + cycle layer.

Cohort (founder / research)

- **Interceptive profiles by sport, age and maturation** — treated as research / hypothesis-generating, not diagnostic; any index needs sport-specific validation first (Welsh).
- **Instruments:** a youth self-report scale (e.g. MAIA-Y) as a periodic trait baseline; the daily loop as the state stream.
- **Governance:** minors' data — guardian consent, athlete-owned by default, clear collection and access. This also keeps reporting honest.

Research in real time — the closed loop

The individual model and the collective evidence base feed each other. Each girl's consented, de-identified pattern joins thousands of others; because our data is cycle-aligned, load-linked and pain-located, it can generate new evidence in precisely the area both Temm and Beato call the biggest gap in sport science — how the menstrual cycle interacts with training load and pain in female (youth) athletes. That discovery then flows back to sharpen every athlete's personal model.

AGGREGATE Thousands of consented, de-identified, cycle-aligned records	→	DISCOVER New evidence on cycle, load and pain	→	FEED BACK A sharper personal model for every athlete
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CLOSED LOOP — discovery feeds her model; her model feeds discovery.

Ethics make it a virtuous loop, not extraction. Participation is opt-in with guardian consent for minors, data is de-identified, and the girl *benefits directly* through the feedback — she is not just a data source. And humility holds: aggregated discovery is hypothesis-generating and sharpens priors; it does not become a clinical claim until independently validated.

10 · Guardrails & safeguarding

- **Calibration, not offloading.** Success = she still reads herself well without the device; run periodic device-off reflections.
- **No placebo scores.** No red 'DANGER' numbers; show gaps as learning; protect a neutral, low-arousal space before competition.
- **Pain with nuance.** Train protective-vs-adaptive discrimination, not hypervigilance; a concerning pain report is never overridden by a green device reading.
- **Privacy, agency, body-positive & trauma-aware.** Menstrual and pain data are hers; coach-sharing is opt-in; language is curious, never surveillance.
- **Age-scaled & reliability-checked.** Simpler body regions for 11–13; establish test–retest reliability per age band before trusting the data.
- **Clinician oversight.** Red-flag routing, menstrual/RED-S pathways and any health guidance are signed off by a qualified sports-medicine clinician.

11 · Outputs & communicating with adults

For a minor the app **supports** the push/rest/recover decision — it never prescribes. Two outputs carry it:

- **A gentle, non-diagnostic nudge** tied to her own fingerprint and load.
- **A shareable weekly 'body-story'** — a simple visual of what her body has been signalling — that she can hand to a coach, parent or physio, building the communicate-to-an-adult skill.

Red flags (gives-way, night pain, focal bone pain, swollen/hot, persistent numbness) escalate to a human immediately, never buried in a trend.

12 · What we deliberately leave out (and why)

- **Food, hunger, fullness, calories** — age-related food-sensitivity risk; energy + thirst are the safe proxies.
- **Acute:chronic workload as a core engine** — weak prediction ($AUC \leq 0.60$) and unstudied in female youth (Tenn); kept only as a soft load-spike nudge.
- **Confident z-score thresholds** — 'smallest worthwhile change' undefined for this group; stay within-athlete and descriptive.
- **Menstrual-phase precision-normalisation** — unreliable in irregular adolescent cycles; use cycle as context only.
- **HRV as a backbone** — too individually variable and device-dependent; optional Tier-2 layer only.

13 · Open decisions & next step

- **Coach-visibility default (needs your call).** Safeguarding wants adults to see red flags; honest reporting wants the data to feel hers. Current default: athlete-owned, opt-in sharing — revisit if deployed through clubs.
- **Validation plan.** Establish body-map reliability per age band and validate the prediction-error/fingerprint measures within-athlete against real outcomes before any claim.

- **Next step:** storyboard the pre- and post-check screens — prompts, the ≤60–90 s tap-flow through the body map, and how surprise and prediction error are captured and shown back.

Design specification, not medical advice. All clinical thresholds, red-flag routing and menstrual/RED-S pathways must be reviewed and signed off by a qualified sports-medicine clinician, and instrument reliability established per age band, before use with minors.

Evidence base: Welsh PS, Seshadri D, Kiely J, Khalsa S. Interoceptive Intelligence in Sports (preprint, 2026). · Temm DA, Standing RJ, Best R. Training, Wellbeing and Recovery Load Monitoring in Female Youth Athletes. IJERPH 2022;19:11463. · Beato M, Madsen EE, Clubb J, Emmonds S, Krstrup P. Monitoring Readiness to Train and Perform in Female Football. IJSPP 2024;19:223–231. · Supporting: OSTRC / Youth Overuse Injury Questionnaire; Adolescent Pediatric Pain Tool; Foxen-Craft et al. Eur J Pain 2019; MAIA-Y.